

Week 2

Introductions & Syllabus

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Introduction

- ▶ Dr. Jonathan A. Pedroza (JP; he/him)
 - ▶ Please call me JP
 - ▶ Everyone *only* gets one “professor/instructor/doctor Pedroza”
- ▶ I'm a first-generation college student
 - ▶ Only one in my family to get a PhD
- ▶ Inspiration: My undergraduate advisor, Dr. Alejandro Morales
- ▶ Hobbies: Roasting coffee, cooking, gardening, and fishing
- ▶ Useless skill: Can make myself cry on cue

Background

- ▶ PhD in Prevention Science from University of Oregon
 - ▶ Identify as a Prevention Scientist
- ▶ R user
- ▶ My academic journey is a roller coaster
 - ▶ Cognition & neuroimaging
 - ▶ Latina/o/x health disparities
 - ▶ Adolescent substance use
 - ▶ Neighborhood composition
 - ▶ Educational assessments
 - ▶ Data Science education

Do you have the credentials to even teach this class?

- ▶ I've taught multiple courses
 - ▶ Statistics
 - ▶ Experimental research design
 - ▶ Seminar research group for McNair scholars
- ▶ Past experience as a data scientist for an educational research center
 - ▶ What was my most used skill?
- ▶ You all have awesome technical skills
 - ▶ Focus on:
 - ▶ Soft skills
 - ▶ Feasibility
 - ▶ Research skills

Introductions

- ▶ What is your name?
 - ▶ What are your pronouns?
- ▶ Where are you from?
- ▶ What is your domain emphasis?
- ▶ Tell us about someone who has been an inspiration to you.
- ▶ Tell us about a hobby or interest of yours.
- ▶ Tell us about a useless skill that you have.
- ▶ What are you most excited for in this class? What is most terrifying?

Data H195 Format

- ▶ 2 semester sequence to conduct an independent data science project
 - ▶ Fall: seminar on research skills, advisor relationship, data access, and a feasible project plan
 - ▶ Spring: complete and advance your thesis, find a letter writer/reference, and build your network

What do I expect from you?

- ▶ Attendance
 - ▶ Come to class
 - ▶ Be ready to talk as a larger group and in smaller groups
 - ▶ Provide meaningful feedback for peers
- ▶ Be respectful and patient with one another
- ▶ Be ready to keep yourself on task for the assignments in this course
 - ▶ Plus whatever your advisor needs for your project
 - ▶ CITI training
 - ▶ Data collection techniques
 - ▶ New analytic techniques
 - ▶ Software

Let's get to the course

Independent Projects

- ▶ What is needed of you
 - ▶ Hustle!
 - ▶ Keep yourself on task
 - ▶ Provide initiative
- ▶ Where I come in
 - ▶ Help with the research process
 - ▶ Accountability & deadlines
 - ▶ Outlines for work

Course Components

- ▶ Finding an advisor
- ▶ Learn fundamentals of conducting research
 - ▶ Literature reviews
 - ▶ Research questions/hypotheses
 - ▶ Theory
 - ▶ Keeping yourself accountable (e.g., timelines)
- ▶ Additional skills
 - ▶ Accountability
 - ▶ Writing
 - ▶ Presenting
 - ▶ Reproducibility

Generative AI Policy

- ▶ AI is allowed for
 - ▶ Brainstorming & idea generation
 - ▶ generate initial ideas
 - ▶ Coding assistance
 - ▶ Grammar checks
- ▶ AI is not allowed for
 - ▶ Generating any writing at all
 - ▶ Summarizing any reading, assignment, or instructions

Assignments

- ▶ Assignments will help build parts of your project thesis
- ▶ Grading will be on bcourses
- ▶ Peer feedback for presentations
- ▶ You will do fine in this class as long as you are putting in effort and submitting assignments
- ▶ This course is not about the grade. It is about you learning how to conduct independent research

Grade Breakdown

Assignment	% of Grade	Points
In-class participation	5%	5
Literature review/annotated bibliographies	5%	5
Advisor check in	5%	5
Timeline	5%	5
Methodology proposal	10%	10
Preliminary proposal presentation	10%	10
Data documentation	10%	10
Final proposal presentation	20%	20
Final proposal	30%	30

Annotated Bibliographies

- ▶ Read literature on your topic of interest
- ▶ Gather key information in either short paragraphs or spreadsheets
 - ▶ Purpose of the statement
 - ▶ Guiding theory
 - ▶ Research questions/hypotheses
 - ▶ Specific methods
 - ▶ Main findings

Submit RQs/Hypotheses

- ▶ Develop RQs and/or hypotheses about your topic of interest
 - ▶ Guided by previous literature
- ▶ Focus will be on
 - ▶ Whether the RQs are feasible in a year
 - ▶ If your data is capable of answering the RQs
 - ▶ If it aligns with your advisor's expertise

Get RQs Approved

- ▶ Share your RQs with your advisor
- ▶ Get RQs approved over email
 - ▶ Submit screenshot of approval/conversation for credit

Submit Timeline

- ▶ Breakdown of weekly activities
- ▶ Useful for keeping yourself accountable
- ▶ You will share this with your advisor
 - ▶ No need for any proof here

Methodology Proposal

- ▶ You will write about your data
 - ▶ Your data source
 - ▶ Explanations of your sample
 - ▶ Steps of data manipulation, processing, and documentation (i.e.,
, missing data, cleaning)
 - ▶ How was data collected
- ▶ Analytic approaches
 - ▶ Why specific models (literature backed)
- ▶ Metrics used and why

Preliminary Proposal Presentation & Feedback

- ▶ Small group presentations
 - ▶ Presenting to others outside of your field
- ▶ You are all expected to give feedback to everyone in your group

Create Repository for Data

- ▶ Depending on the type of data you have
 - ▶ You'll either use Box, a qualitative data service, or GitHub

Final Proposal Presentation & Feedback

- ▶ Finalized presentation on your proposal that you will take with you into the spring semester
- ▶ You'll give feedback to every other student

Final Proposal

- ▶ An accumulation of all of your work in the form of an academic paper

Questions???